

4.2. Reward model training and regularized objective.

- (a) A reward model outputs the following scores for three preference pairs:

Pair	$r(x, y^+)$	$r(x, y^-)$
1	2.0	1.0
2	0.5	1.5
3	3.0	0.0

Compute the Bradley-Terry loss for each pair: $\ell_i = \log \sigma(r(x, y_i^+) - r(x, y_i^-))$, where $\sigma(z) = 1/(1 + e^{-z})$. Then compute the average loss $\bar{\ell} = -\frac{1}{3} \sum_i \ell_i$.

- (b) Write down the regularized RLHF reward maximization objective $\mathcal{J}(\theta)$ for a policy π_θ . For a single generated response y with reward $r(x, y) = 1.5$ and $\log(\pi_\theta(y | x)/\pi_{\text{ref}}(y | x)) = 0.3$, compute the regularized reward $r_s(x, y)$ with $\beta = 0.1$. Interpret whether this response is beneficial to keep or suppress.

Question 4.2 [8 points] [\[Mathematical\]](#) **Regularized Reward Maximization.** The RLHF training objective for a policy π_θ is the KL-regularized reward:

$$\mathcal{J}(\theta) = \mathbb{E}_{\pi_\theta(y|x)} \left[r(x, y) - \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)} \right].$$

- (a) (2 pts) Explain the role of the KL penalty term $\beta \log(\pi_\theta/\pi_{\text{ref}})$ and what happens when $\beta \rightarrow 0$ and $\beta \rightarrow \infty$.
- (b) (3 pts) In PPO-Clip, the importance-weighted gradient update uses:

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}[\min(\rho_t A_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) A_t)],$$

where $\rho_t = \pi_\theta(a_t|s_t)/\pi_k(a_t|s_t)$ is the importance ratio and A_t is the advantage. Explain what the `clip` operation achieves. Consider the case $A_t > 0$: what is the effective gradient when $\rho_t > 1 + \epsilon$?

- (c) (3 pts) Write down the advantage function $A(s_t, a_t)$ in terms of the Q-function $Q(s_t, a_t)$ and value function $V(s_t)$. Explain intuitively why using A instead of Q reduces variance in policy gradient updates.